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Management of Upper Urinary Tract Obstruction

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Keywords

diuretic renography; noncontrast CT; renal ultrasound; retrocaval ureter; UPJO; ureteral stricture disease; ureteroenteric anastomotic stricture and retroperitoneal fibrosis

Questions

1. Ureteropelvic junction (UPJ) obstruction in the neonate is most frequently found as a result of:
    a. maternal-fetal ultrasonography.
    b. voiding cystourethrography.
    c. diuretic renography.
    d. abdominal radiography.
    e. physical examination.
2. Which study is diagnostic for functional obstruction at the UPJ?
    a. Retrograde pyelography
    b. Three-dimensional helical computed tomography (CT)
    c. Diuretic renography
    d. Renal ultrasound
    e. Renal angiography
3. A 62-year-old man presents with left flank pain. CT urography reveals delayed excretion and hydronephrosis to the level of a 2.5-

cm calculus at the UPJ. Percutaneous stone extraction is accomplished without difficulty, but a postextraction nephrostogram reveals hydronephrosis to the level of the UPJ without residual stone. A follow-up nephrostogram 1 week later is unchanged. The best next step is:

a. removal of the nephrostomy tube.
b. diuretic renography.
c. CT angiography.
d. antegrade endopyelotomy.
e. Whitaker pressure-perfusion test.

4. A 27-year-old woman has right flank pain, and her diuretic renography reveals UPJ obstruction and a differential renal function of 75:25 (L:R). The next best step is:

a. CT angiography.
b. stent placement.
c. endopyelotomy.
d. laparoscopic pyeloplasty.
e. laparoscopic nephrectomy.

5. The highest failure rate in treating UPJ obstruction is associated with:

a. antegrade endopyelotomy.
b. retrograde ureteroscopic endopyelotomy.
c. balloon dilation.
d. pyeloplasty.
e. cautery balloon incision.

6. The most appropriate location for endoscopic incision of a proximal ureteral stricture is:

a. lateral.
b. anterior.
c. medial.
d. posterior.
e. anterolateral.

7. The best treatment option for a patient with a functional left ureteroenteric anastomotic stricture is:

a. metallic stent.
b. balloon dilation.

c. laser endoureterotomy.
d. cautery wire balloon incision.
e. open repair.

8. The most common cause of retroperitoneal fibrosis is:
   a. methysergide.
   b. infection.
   c. lymphoma.
   d. breast cancer.
   e. immune-mediated aortitis.
9. Retrocaval ureter results from:
   a. persistence of posterior cardinal veins.
   b. persistence of anterior cardinal veins.
   c. duplication of inferior vena cava.
   d. aberrance of lumbar veins.
   e. retroaortic renal veins.
10. Transperitoneal laparoscopic pyeloplasty:
   a. is used rarely compared with the retroperitoneal approach.
   b. does not require water-tight, tension-free anastomosis.
   c. provides more working space than in the retroperitoneal approach.
   d. provides unfamiliar anatomy.
   e. does not require an external surgical drain.
11. Surgical repair of UPJ obstruction requires:
   a. a funnel-shaped transition between the renal pelvis and ureter.
   b. dependent drainage.
   c. water-tight anastomosis.
   d. tension-free anastomosis.
   e. all of the above.
12. Contraindications for transureteroureterostomy include a history of:
   a. retroperitoneal fibrosis.
   b. urothelial malignancy.
   c. nephrolithiasis.
   d. a, b, and c.
   e. b and c.
13. A 25-year-old man presents with right flank pain. He underwent a laparoscopic pyeloplasty, which failed within 1 year. Consequently,

he underwent failed endopyelotomy. A CT scan shows a small, intrarenal pelvis and moderate cortical loss in the right kidney with a normal-appearing left kidney. A renogram reveals 35% differential function on the affected side, and a diuretic study demonstrates functional obstruction (>30 minutes). The next step is:

a. chronic internal ureteral stent.
b. ileal ureter.
c. Davis intubated ureterotomy.
d. ureterocalicostomy.
e. renal autotransplantation.

14. Spiral flap procedures for UPJ obstruction are used:
    a. to bridge a shorter-length stenosis.
    b. to treat crossing vessels.
    c. to bridge a longer-length stenosis.
    d. for a small, intrarenal pelvis.
    e. only in the presence of greater than 30% ipsilateral renal function.
15. Foley Y-V-plasty is a suitable approach when encountering:
    a. high ureteral insertion.
    b. small intrarenal pelvis.
    c. anterior crossing vessel.
    d. duplication of collecting system.
    e. redundant renal pelvis.
16. Which type of pyeloplasty is suitable when there is an aberrant crossing vessel?
    a. Foley Y-V-plasty
    b. Culp-DeWeerd spiral flap
    c. Dismembered pyeloplasty
    d. Scardino-Prince vertical flap
    e. Ligation and transection of the crossing vessel
17. Ileal ureter can be performed when the patient has:
    a. renal insufficiency (serum creatinine > 2 mg/dL).
    b. inflammatory bowel disease.
    c. bladder dysfunction.
    d. radiation enteritis.

e. small intrarenal pelvis.

18. A 55-year-old woman underwent left transperitoneal laparoscopic dismembered pyeloplasty over an internal ureteral stent. An abdominal drain was placed at surgery, and there was minimal drain output during the first 24 hours after surgery. Within 3 hours after Foley catheter removal, the patient's nurse noted a significant amount of fluid coming out of the drain site. The next step is to:
    a. change dressings frequently and continue observation.
    b. replace the urethral catheter.
    c. restrict fluid intake.
    d. remove the surgical drain.
    e. change the ureteral stent.
19. In performing a psoas hitch, additional bladder mobility can be achieved by transection of the:
    a. contralateral superior vesical artery.
    b. ipsilateral inferior vesical artery.
    c. contralateral inferior vesical artery.
    d. ipsilateral superior vesical artery.
    e. ipsilateral gonadal artery.
20. A 40-year-old woman with a history of hypertension and recurrent nephrolithiasis presents with a 5-cm proximal right ureteral stricture following an iatrogenic injury in a recent abdominal surgery. She has had an indwelling right nephrostomy tube for more than 6 months. Her baseline serum creatinine is 2.5 mg/dL. Renal scan shows split function of 65% in the right kidney. Her bladder capacity is found to be less than 300 mL. The next step is:
    a. ureteroureterostomy.
    b. Boari flap.
    c. transureteroureterostomy.
    d. ileal ureteral substitution.
    e. autotransplantation.
21. During a psoas hitch, the structure particularly susceptible to injury is the:
    a. obturator nerve.
    b. iliohypogastric nerve.
    c. ilioinguinal nerve.

d. sacral nerve.
e. genitofemoral nerve.

22. The technique that does not require normal bladder capacity, drainage, and function is the:
    a. ileal ureteral substitution.
    b. psoas hitch.
    c. ureteroneocystostomy.
    d. endoscopic incision of transmural ureter.
    e. Boari flap.

Imaging

1. See Fig. 89.1. A 72-year-old man with malaise has this CT scan. The serum creatinine is mildly elevated. What is the best diagnosis?
    a. Retroperitoneal fibrosis
    b. Retroperitoneal hematoma



FIG. 89.1

    c. Tuberculosis
    d. Retroperitoneal sarcoma

e. Perianeurysmal fibrosis

Answers

1. **a. Maternal-fetal ultrasonography.** The current widespread use of maternal ultrasonography has led to a dramatic increase in the number of asymptomatic newborns who are diagnosed with hydronephrosis, many of whom are subsequently found to have UPJ obstruction.
2. **c. Diuretic renography.** Provocative testing with a diuretic urogram may allow accurate diagnosis of UPJ obstruction. Renal ultrasound, CT scan, and retrograde pyelogram give anatomic assessments of the UPJ without quantitatively assessing urinary drainage and function.
3. **e. Whitaker pressure-perfusion test.** When doubt remains as to the clinical significance of a dilated collecting system, placement of percutaneous nephrostomy allows access for pressure perfusion studies. In the pressure perfusion test, first described by Whitaker in 1973 and then modified in 1978, the renal pelvis is perfused with normal saline or dilute radiographic contrast solution, and the pressure gradient across the presumed area of obstruction is determined. Renal pelvic pressures in excess of 15 to 22 cm H_2O are highly suggestive of a functional obstruction. Although diuretic renography is useful for diagnosis as well, the Whitaker test is ideal for this situation because a nephrostomy tube is already in situ.
4. **b. Laparoscopic pyeloplasty.** Evidence from several investigators indicates that crossing vessels lower the success rate of endopyelotomy. When such patients were culled from the pool of candidates available for treatment of UPJ obstruction, endopyelotomy success rates improved in most studies. A CT angiography would be necessary to assess this. However, laparoscopic pyeloplasty would be a straightforward minimally invasive option for this young patient. The kidney has too much function to remove at this stage.

5. **c. Balloon dilation. McClinton reported long-term follow-up data on balloon dilation of the UPJ, finding a success rate of only 42%, which was significantly lower than the initial publications would indicate.**
6. **a. Lateral. Proximal ureteral strictures are incised laterally, similar to UPJ strictures. Posterior incision is offered to UPJ obstruction patients who have failed open pyeloplasty. Distal strictures are incised anteriorly, as are strictures of the middle ureter.**
7. **e. Open repair.** Several studies have linked poor outcomes with endoscopic management of left ureteroenteric strictures. This may be a result of diminished blood flow to the ureter because the left ureter requires more mobilization than the right side at the time of diversion. Although metallic stents show promise in limited studies, using open repair, reports demonstrate an 80% success.
8. **e. Immune-mediated aortitis.** Growing evidence indicates that the majority of cases of retroperitoneal fibrosis are, in fact, immune-mediated aortitis. Regardless, the other conditions are relatively rare causes of retroperitoneal fibrosis.
9. **a. Persistence of posterior cardinal veins.** Retrocaval ureter results from the persistence of the posterior cardinal veins.
10. **c. Provides more working space than that in the retroperitoneal approach.** Transperitoneal laparoscopic pyeloplasty provides a larger working space relative to a retroperitoneoscopic approach. Together with more familiar anatomy, the transperitoneal approach is used most commonly in the laparoscopic urologic community to date.
11. **e. All of the above. For any surgical repair of UPJ obstruction, the resultant anastomosis should be widely patent and completed in a watertight fashion without tension. In addition, the reconstructed UPJ should allow a funnel-shaped transition between the pelvis and the ureter that is in a position of dependent drainage.**
12. **d. a, b, and c. Relative contraindications include history of nephrolithiasis, retroperitoneal fibrosis, urothelial malignancy, chronic pyelonephritis, and abdominopelvic radiation.**

13. **d. Ureterocalicostomy.** Direct anastomosis of the proximal ureter to the lower calyceal system is a well-accepted salvage technique for the failed pyeloplasty and small renal pelvis.
14. **c. To bridge a longer-length stenosis.** Flap procedures can be useful in situations involving a relatively long segment of ureteral narrowing or stricture. Of the various flap procedures, a spiral flap can bridge a strictured or narrow area of longer length. The flap procedures are not appropriate in the setting of crossing vessels.
15. **a. High ureteral insertion.** The Foley Y-V-plasty is designed for repair of a UPJ obstruction secondary to a high ureteral insertion. It is specifically contraindicated when transposition of lower pole vessels is necessary. In situations requiring concomitant reduction of redundant renal pelvis, this technique is also of little value.
16. **c. Dismembered pyeloplasty.** In the presence of crossing aberrant or accessory lower pole renal vessels associated with UPJ obstruction, a dismembered pyeloplasty is the only method to allow transposition of the UPJ in relation to these vessels.
17. **e. Small intrarenal pelvis.** In ileal segment usage, a small intrarenal pelvis is not contraindicated and an ileocalycostomy can be performed successfully.
18. **b. Replace the urethral catheter.** If the drain output increases following the removal of the Foley catheter, the catheter should be replaced for several days to avoid vesicoureteral reflux up the stent in the operated ureter and decrease urinary extravasation.
19. **a. Contralateral superior vesical artery.** In psoas hitch, transection of the contralateral superior vesical artery can be helpful to bridge the gap to the ipsilateral ureteral end, thereby achieving tension-free anastomosis.
20. **e. Autotransplantation.** Ureteroureterostomy is inappropriate for a 5-cm upper ureteral defect. Boari flap is inappropriate for a small bladder capacity. Transureteroureterostomy is contraindicated in the patient with a history of recurrent nephrolithiasis. Ileal ureter is contraindicated in the presence of elevated serum creatinine above 2 mg/dL. Autotransplant is appropriate for this particular patient.
21. **e. Genitofemoral nerve.** The genitofemoral nerve courses over the psoas muscle.

22. **d. Endoscopic incision of transmural ureter.** Normal bladder function without significant outlet obstruction is crucial to the success of ileal ureteral substitution, psoas hitch, Boari flap, and ureteroneocystostomy.

Imaging

1. **a. Retroperitoneal fibrosis.** There is increased soft tissue in the retroperitoneum, which obscures and effaces the planes between the inferior vena cava and the aorta. Tuberculosis causes calcification and stricturing in the kidneys and collecting systems, and tuberculous iliopsoas abscess extends along the iliopsoas muscles. Retroperitoneal hematoma and sarcoma are not centered solely around the aorta and the inferior vena cava. In perianeurysmal fibrosis, a retroperitoneal fibrosis-like picture occurs in association with an abdominal aortic aneurysm; the aorta is of normal caliber in this case.

Chapter Review

1. Intrinsic UPJ obstruction is a result of an aperistaltic segment in which the normal spiral arrangement of the muscle bundles is replaced by longitudinal muscle bundles and fibrous tissue.
2. A crossing vessel has the most detrimental effect on the success of an endopyelotomy.
3. UPJ obstruction may coexist with vesicoureteral reflux.
4. A multicystic kidney is distinguished from a UPJ obstruction on ultrasound by the “cyst” being connected in hydronephrosis as opposed to being distinct in a multicystic dysplastic kidney.
5. A Whittaker test is performed by perfusing the renal pelvis at 10 mL/min. Pressures less than 15 cm H_2O suggest a nonobstructed system. Pressures greater than 22 cm H_2O suggest an obstructed system, and pressures between the two are indeterminate.
6. The indications for repair of a UPJ include symptoms, impairment of renal function, stones, infection, and hypertension.

7. In neonates, unilateral hydronephrosis when carefully followed results in 7% of patients requiring a pyeloplasty.
8. Generally, kidneys with less than 15% function are not salvageable in adult patients.
9. A long segment stricture (>2 cm) is generally not successfully managed by the endopyelotomy method. An endopyelotomy cannot be performed safely by any route until access across the UPJ is established.
10. The majority of endopyelotomy failures occur within the first year. Success rates for endopyelotomy in properly selected patients range between 60% and 80%.
11. High-grade hydroureteronephrosis and crossing vessels have a detrimental effect on the success rate of endopyelotomy.
12. When bleeding occurs following an endopyelotomy, one should have a low threshold to proceed to angiography to thrombose the severed vessel.
13. Seventy percent of failures following laparoscopic pyeloplasty occur in the first 2 years.
14. When repairing a retrocaval ureter, the ureter is transected and relocated ventral to the vena cava.
15. Lower ureteral strictures are incised in an anterior medial direction; upper ureteral strictures are incised in a lateral or posterior lateral direction.
16. With ureteral strictures, one must always rule out malignancy.
17. There is no significant difference in preserving renal function in the adult when reimplanting the ureter into the bladder by either a refluxing or antirefluxing method.
18. Most patients with long-term urinary conduits will have an element of hydronephrosis that is not secondary to obstruction.
19. Retroperitoneal fibrosis secondary to malignancy is often indistinguishable from idiopathic retroperitoneal fibrosis and can be identified only with appropriate biopsy that identifies islands of tumor cells.
20. The initial management of retroperitoneal fibrosis is generally with steroids. Steroids are more likely to be beneficial if there is evidence of active inflammation as indicated by an elevated erythrocyte

sedimentation rate, leukocytosis, and infiltration of lymphocytes on biopsy.

21. In addition to steroids, azathioprine, cyclophosphamide, cyclosporine, colchicine, and tamoxifen have been used to treat retroperitoneal fibrosis with some success.
22. Generally, 25% renal function is required to keep a repair of the UPJ or ureter open.
23. For any surgical repair of UPJ obstruction, the resultant anastomosis should be widely patent and completed in a watertight fashion without tension. In addition, the reconstructed UPJ should allow a funnel-shaped transition between the pelvis and the ureter that is in a position of dependent drainage.

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Upper Urinary Tract Trauma

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Keywords

kidney; renal trauma; upper urinary tract

Questions

1. Which of the following are mechanisms of injury or physical exam findings concerning for a potential blunt renal injury?
    a. Significant blow to the flank
    b. Lower rib fractures
    c. Significant flank ecchymoses
    d. A fall from a third-story window
    e. All of the above
2. What is the best option for repair of midureteral transection after a stab wound, in a stable patient?
    a. Ureteroureterostomy
    b. Transureteroureterostomy
    c. Boari flap
    d. Nephrectomy
    e. Cutaneous ureterostomy
3. When ureteroureterostomy is performed, which of the following is required?
    a. Postoperative interposition of omentum
    b. Postoperative nephrostomy tube

c. Spatulated, watertight repair over a ureteral stent
d. Nonabsorbable sutures
e. Intraperitonealization of the ureteral anastomosis

4. Which maneuver is a common cause of ureteral injury during stone basketing?
   a. Ureteroscopy without dilating the ureteral orifice first
   b. Ureteroscopy in nondilated systems
   c. Use of the holmium laser
   d. Pulsatile saline irrigation to assist visualization
   e. Persistence in stone basketing attempts in the face of a ureteral tear
5. Which of the following is a contraindication to transureteroureterostomy for repair of significant lower ureter injury?
   a. History of urolithiasis
   b. History of ureteral trauma
   c. Morbid obesity
   d. Neurogenic bladder
   e. Thoracic spine fracture
6. What is the best treatment for a short, mid ureter ureteral contusion caused by a high-velocity bullet?
   a. Observation
   b. Ureteral stent placement
   c. Transureteroureterostomy
   d. Excision and ureteroureterostomy
   e. Oversew the contusion with healthy ureteral tissue
7. Which imaging technique is most useful for detecting ureteral injuries in the stable trauma patient?
   a. Computed tomography (CT) of the abdomen/pelvis without use of contrast material
   b. CT of the abdomen and pelvis with IV contrast with immediate images only
   c. CT of the abdomen and pelvis with IV contrast with immediate and delayed imaging (10 to 15 minutes)
   d. IVP
   e. Diuretic (Lasix) renography

8. Which of the following statements is TRUE about ureteral injuries during laparoscopy?
    a. The total number of injuries has stayed steady over the years.
    b. Surgery for endometriosis greatly increases the risk.
    c. Bipolar cautery use during tubal ligation eliminates risk.
    d. Most ureteral injuries are recognized immediately.
    e. Indigo carmine dye injection eliminates the risk of injury.

Imaging

1. A 22-year-old woman has a CT scan (Fig. 90.1) after a motor vehicle accident. Her vital signs are stable, and there are no other significant injuries.

The next step is:

a. open surgical repair of the kidney.
b. delayed imaging to evaluate the collecting system.
c. left nephrectomy to avoid future complications.
d. renal artery embolization.
e. intravenous urogram.

2. A 42-year-old man had a CT scan (Fig. 90.2) 36 hours after a motor vehicle accident. The most likely diagnosis is:
    a. ureteral injury.
    b. segmental renal artery transection.
    c. main renal artery occlusion.
    d. ureteropelvic junction disruption.
    e. traumatic renal vein occlusion.

Answers

1. **e. All of the above.**



FIG. 90.1

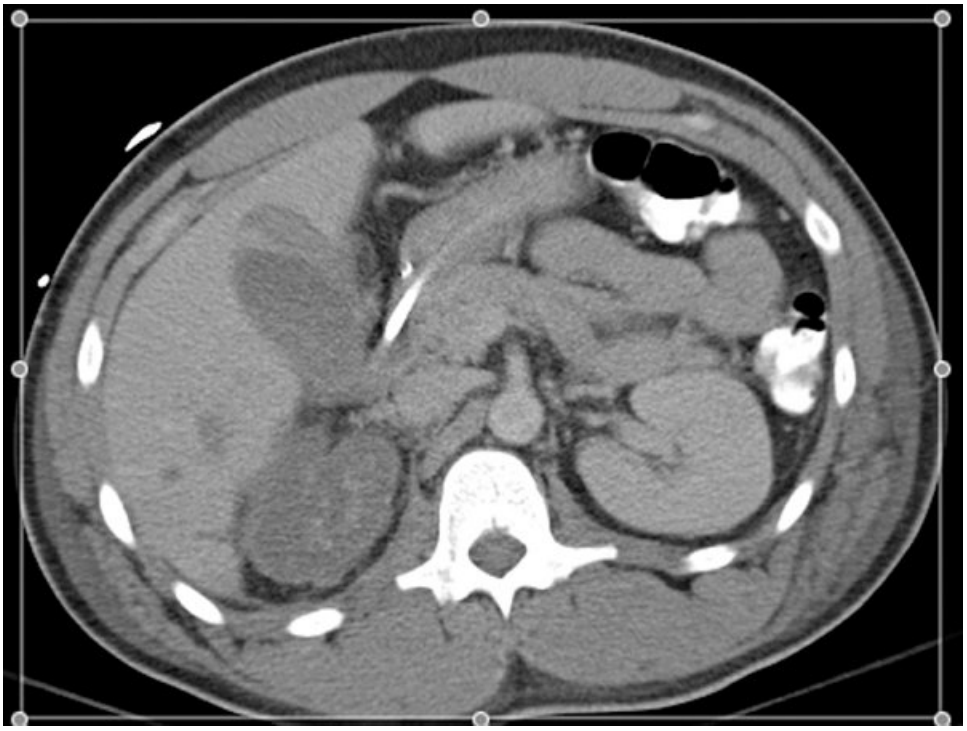


FIG. 90.2

2. **a. Ureteroureterostomy.** Ureteroureterostomy, so-called end-to-end repair in injuries to the upper two-thirds of the ureter, is common (up to 32% of one large series) and has a reported success rate as high as 90%.
3. **c. Spatulated, watertight repair over a ureteral stent.** A proper ureteral repair entails a spatulated, tension-free, stented, watertight anastomosis and a retroperitoneal closed suction drain.
4. **e. Persistence in stone basketing attempts in the face of a ureteral tear.** One factor cited as a cause of injury was the persistence in stone basketing attempts after recognition of a ureteral tear. Current recommendations are to stop and place a ureteral stent.
5. **a. History of urolithiasis.** TUU is contraindicated in patients with a history of kidney stones because a distal ureteral obstructing stone can put then put both kidneys at risk.
6. **d. Excision and ureteroureterostomy.** High-velocity gunshot wounds to the ureter result in blast injury and thus often heal with stricture or break down from ureteral necrosis. Such contusions

should be treated with excision and ureteroureterostomy. Low-velocity bullet ureter contusions can be safely managed by ureteral stent placement.

7. **c. CT of the abdomen and pelvis with IV contrast with immediate and delayed imaging (10 to 15 minutes).** Because CT scanners can obtain early phase images before intravenous contrast dye is excreted in the urine, delayed images must be obtained (10 to 15 minutes after contrast material injection) to allow sufficient time for contrast to be excreted by the kidney and into the collecting system, renal pelvis, and down the ureter.
8. **b. Surgery for endometriosis greatly increases the risk.** A large percentage of ureteral injuries after gynecologic laparoscopy occur during electrosurgical or laser-assisted lysis of endometriosis.

Imaging

1. **b. Delayed imaging to evaluate the collecting system. When blunt renal trauma results in deep parenchymal lacerations, delayed imaging at the time of the CT is essential to evaluate the collecting system for injury.** In a stable patient with this type of injury, neither surgical repair nor angiographic embolization is indicated. Most renal injuries heal well with conservative and expectant management, making options a and c incorrect.
2. **c. Main renal artery occlusion.** There is complete absence of enhancement of the entire right kidney, which is caused by intimal injury and then main renal artery occlusion. Segmental occlusion would cause abnormality only in a portion of the right renal nephrogram. Ureteral injury and UPJ disruption may cause delay in the nephrogram but not complete absence of enhancement.

Chapter Review

1. The best indication of significant urinary system injury is gross hematuria. However, the absence of hematuria does not exclude a significant GU injury.

2. Evaluation of urologic trauma in children differs from adults in that children: (1) are at greater risk for renal trauma, (2) often do not become hypotensive with major blood loss, and (3) have a higher propensity for renal anomalies.
3. Rapid deceleration from a fall from height or a high-velocity impact may result in injuries at points of fixation such as the ureteral pelvic junction and the renal hilum (renal artery intimal disruption).
4. The degree of hematuria and the severity of renal injury are not consistently correlated.
5. Criteria for radiologic imaging include (1) all penetrating trauma, (2) high-impact rapid deceleration trauma, (3) all blunt trauma with gross hematuria, (4) all blunt trauma with microhematuria and hypotension, and (5) pediatric patients with hematuria.
6. Adult patients with microscopic hematuria without shock may be observed without imaging studies.
7. Findings suggestive of a major renal injury on CT include medial laceration, perinephric hematoma size, intravascular contrast extravasation, medial urinary extravasation, and devitalized renal fragments.
8. Nonoperative management for renal injuries is preferred in the hemodynamically stable patient, particularly with grades I to III renal injuries.
9. Exploration of low-velocity renal gunshot wounds is not mandatory in selected cases. Such patients with isolated renal parenchymal lacerations with stable and contained hematomas who are hemodynamically stable, with no intra-abdominal organ injury, may be observed expectantly.
10. Absolute indications for renal exploration are (1) hemodynamic instability with shock, (2) expanding or pulsatile hematoma, (3) suspected renal pedicle avulsion, and (4) uncontained retroperitoneal hematoma.
11. For kidney trauma, perinephric hematoma size, intravascular contrast extravasation, devitalized renal fragments, arterial thrombosis, and collecting system contrast extravasation are relative indications for intervention.

12. “Damage control” is the management technique of immediate control of bleeding, and fecal and urinary leak only, in the unstable patient. Definitive reconstruction is deferred until the patient is stable and fully resuscitated. For ureter injuries this entails ligation and nephrostomy, externalizing, or stenting. At a staged/planned laparotomy, the ureter is later repaired.
13. Hypertension from renal parenchymal injury is usually short lived. Renal vascular injury or compression of the renal parenchyma by a subcapsular hematoma (Page kidney) can also cause high blood pressure.
14. When repairing ureteral injuries, the ureteral tissue should be debrided back to a bleeding edge to remove all traumatized microvascular damaged tissue.
15. Placement of vascular graft aneurysms in proximity of the ureter may cause a periureteral inflammatory reaction and ureteral injury/stenosis.
16. Ureteral injuries to the proximal and mid ureter can often be managed by uretroureterostomy. The gap being bridged is based on ureter mobilization and its elasticity.
17. In the stable trauma patient, distal ureteral injuries should be reimplanted into the bladder. In select cases, a ureteroureterostomy can be considered. If the gap is large, then psoas hitch or Boari bladder flap is the better management.
18. For delayed ureteral injury, the best diagnostic radiographic test (in the stable patient) to confirm injury, is a retrograde urogram.
19. Ureteral transections should be repaired within a week of the injury, or the repair should be delayed for 6 or more weeks.
20. It is prudent to isolate the ureteral repair from other injured organs (such as colon) with omentum or peritoneum.
21. Ureteral injury noted during ureteroscopy should be managed by ureteral stent placement. When a stent cannot be placed or inadequately diverts the urine, then a percutaneous nephrostomy should be placed.
22. Ureteral injury occurring during vascular surgery should be repaired and isolated from the graft with normal tissue such as omentum.

PART X

Urinary Lithiasis and Endourology

OUTLINE

91. Urinary Lithiasis Etiology Epidemiology and Pathogenesis
92. Evaluation and Medical Management of Urinary Lithiasis
93. Strategies for Nonmedical Management of Upper Urinary Tract Calculi
94. Surgical Management for Upper Urinary Tract Calculi
95. Lower Urinary Tract Calculi

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Urinary Lithiasis Etiology Epidemiology and Pathogenesis

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Keywords

calcium oxalate; cystinuria; hypercalciuria; hyperoxaluria; nephrolithiasis; supersaturation; uric acid nephrolithiasis

Questions

1. The ethnic/racial group with the highest prevalence of stone disease is:
    a. African-American
    b. Hispanic
    c. White
    d. Asian
    e. American Indian
2. The geographic area in the United States associated with the highest incidence of calcium oxalate stone disease is the:
    a. northeast.
    b. southeast.
    c. southwest.
    d. west.
    e. northwest.
3. Which of the following conditions is associated with an increased risk of nephrolithiasis?

a. Underweight
b. Metabolic syndrome
c. Rheumatoid arthritis
d. Hypothyroidism
e. Atopy

4. Which of the following occurs when the concentration product of urine is in the metastable range?
   a. Urine is supersaturated.
   b. Homogeneous nucleation occurs.
   c. Solubility product is reduced.
   d. Urinary inhibitors decrease the formation product.
   e. Nucleation never occurs.
5. The process by which nucleation occurs in pure solutions is:
   a. homogeneous nucleation.
   b. heterogeneous nucleation.
   c. epitaxy.
   d. aggregation.
   e. agglomeration.
6. The proteinaceous portion of stones is composed of:
   a. concentric lamination.
   b. protein-crystal complex.
   c. matrix.
   d. nephrocalcin.
   e. osteocalcin.
7. Stone-forming propensity is best described by:
   a. formation product.
   b. ionic activity.
   c. saturation index.
   d. solubility product.
   e. relative saturation ratio.
8. The most common abnormal urinary finding in patients undergoing Roux-en-Y gastric bypass surgery is:
   a. hypercalciuria.
   b. low urine pH.
   c. low urine volume.
   d. hypocitraturia.

e. hyperoxaluria.

9. The vitamin D metabolite that stimulates intestinal calcium absorption is:
   a. 7-dehydrocholesterol.
   b. cholecalciferol.
   c. 25-dihydroxyvitamin D_3.
   d. 1,25-dihydroxyvitamin D_3.
   e. calcitonin.
10. Which of the following factors increases intestinal oxalate absorption?
   a. High dietary calcium intake.
   b. Low dietary calcium intake.
   c. *Oxalobacter formigenes* colonization in the colon.
   d. *Helicobacter pylori* colonization in the stomach.
   e. Irritable bowel syndrome.
11. The primary determinant of urinary citrate excretion is:
   a. acid–base status.
   b. urinary sodium excretion.
   c. citric acid intake.
   d. insulin sensitivity.
   e. urinary calcium excretion.
12. The underlying abnormality of renal hypercalciuria is:
   a. enhanced calcium filtration.
   b. enhanced calcium secretion.
   c. enhanced calcium reabsorption.
   d. primary renal wasting of calcium.
   e. primary renal storage of calcium.
13. Enteric hyperoxaluria occurs as a result of:
   a. excessive intake of oxalate.
   b. reduced excretion of oxalate.
   c. increased dietary fat.
   d. low calcium intake.
   e. fat malabsorption.
14. The most likely mechanism accounting for low urinary pH in uric acid stone formers with type 2 diabetes mellitus is:

a. defective ammoniagenesis.
b. impaired urinary bicarbonate excretion.
c. lactic acidosis.
d. glucosuria.
e. ketoacidosis.

15. In idiopathic calcium oxalate stone formers, Randall plaques originate in the:
    a. basement membrane of the thin loops of Henle.
    b. terminal collecting ducts.
    c. medullary interstitium.
    d. vasa recta.
    e. papillary tip.
16. In calcium oxalate stone formers, Randall plaques are composed of:
    a. calcium oxalate.
    b. brushite.
    c. calcium carbonate.
    d. calcium apatite.
    e. uric acid.
17. Urinary saturation of calcium oxalate is most strongly dependent on:
    a. urinary calcium concentration.
    b. urinary oxalate concentration.
    c. both urinary calcium and oxalate concentrations.
    d. urinary pH.
    e. urinary citrate concentration.
18. *O. formigenes* reduces urinary oxalate by:
    a. reducing intestinal calcium absorption, leading to decreased luminal free oxalate and reduced oxalate absorption.
    b. degrading urinary oxalate in infected urine.
    c. binding oxalate in the intestinal lumen and preventing its reabsorption.
    d. inhibiting the intestinal oxalate transporter.
    e. using oxalate as a substrate in the intestine, thereby reducing intestinal oxalate absorption.
19. Which of the following organisms is most likely to produce urease?
    a. *Staphylococcus aureus*

b. *Escherichia coli*
c. *Streptococcus pneumoniae*
d. *Serratia marcescens*
e. *Chlamydia*

20. The mechanism responsible for type 1 (distal) renal tubular acidosis (RTA) is:
    a. impaired bicarbonate reabsorption in the proximal tubule.
    b. defective H^+-ATPase in the distal tubule that is unable to excrete excess acid.
    c. defective ammoniagenesis.
    d. impaired excretion of nontitratable acids.
    e. hypoaldosteronism.
21. Patients with Lesch-Nyhan syndrome treated with high doses of allopurinol are at risk for formation of stones of which of the following compositions?
    a. Hypoxanthine
    b. Uric acid
    c. Xanthine
    d. 2,8-Dihydroxyadenine
    e. Calcium apatite
22. The etiology of ammonium acid urate stone formation in patients abusing laxatives is:
    a. recurrent infections with urease-producing bacteria.
    b. chronic dehydration and excessive uric acid excretion.
    c. increased ammoniagenesis.
    d. urinary phosphate deficiency and intracellular acidosis.
    e. chronic dehydration, intracellular acidosis, and low urinary sodium.
23. The primary mechanism of action of citrate in preventing stone formation is:
    a. reducing urinary calcium excretion.
    b. reducing urinary oxalate excretion.
    c. complexing calcium in urine.
    d. complexing oxalate in urine.
    e. complexing phosphate in urine.

24. Type 1 (distal) RTA is characterized by which abnormality?
    a. Hyperkalemia
    b. Hypochloremia
    c. Alkalosis
    d. Hypercitraturia
    e. Hypokalemia
25. The primary defect in type 2 (proximal) RTA is failure of bicarbonate reabsorption in the:
    a. glomerulus.
    b. proximal tubule.
    c. loop of Henle.
    d. distal tubule.
    e. collecting duct.
26. The most common abnormality identified in patients with uric acid stones is:
    a. acidic urine.
    b. alkaline urine.
    c. low uric acid concentration.
    d. high uric acid concentration.
    e. distal renal tubular acidosis.
27. Carbonic anhydrase inhibitors are associated with formation of stones composed of:
    a. calcium oxalate.
    b. calcium phosphate.
    c. struvite.
    d. cystine.
    e. uric acid.
28. Which of the following physiologic changes occurs in the kidney during pregnancy?
    a. Decreased uric acid excretion
    b. Decreased citrate excretion
    c. Increased calcium excretion
    d. Decreased glomerular filtration rate (GFR)
    e. Increased magnesium excretion
29. Which of the following substances acts as a promoter of calcium oxalate stone formation?

a. Matrix-Gla protein
b. Bikunin
c. Citrate
d. Pyrophosphate
e. Uric acid

30. Which of the following amino acids is subject to impaired renal reabsorption in cystinuria?
    a. Arginine
    b. Glycine
    c. Alanine
    d. Leucine
    e. Methionine

Answers

1. **c. White.**The highest prevalence of stone disease in both men and women occurs in whites. In men the lowest prevalence occurs in African-Americans, whereas Asian women have been found to have the lowest prevalence in one series.
2. **b. Southeast.** According to hospital discharge rates among U.S. veterans, calcium oxalate stone disease is most prevalent in the southeast.
3. **b. Metabolic syndrome.** Metabolic syndrome is characterized by obesity, hypertension, dyslipidemia, and insulin resistance, and is associated with an increased risk of kidney stone formation. Although impaired ammoniagenesis accounts for an increased risk of uric acid stones due to low urine pH, association with stone disease in general has been demonstrated. None of the other conditions has been associated with stone disease.
4. **a. Urine is supersaturated.**
5. **a. Homogeneous nucleation.** The process by which nuclei form in pure solutions is called *homogeneous nucleation.*
6. **c. Matrix.** Depending on their type, kidney stones contain between 2.5% and 65% of noncrystalline material or matrix. Extensive

investigations have characterized matrix as a derivative of several of the mucoproteins of urine and serum.

7. **c. Saturation index.** The state of saturation of the urine with respect to particular stone-forming salts indicates the stone-forming propensity of the urine. The state of saturation is determined by pH and the ionic strength of the major ions in solution. Relative saturation ratio, determined by the EQUIL 2 computer program, has been the standard for determining stone-forming propensity. However, the newer *JESS* computer program takes into account several soluble complexes not recognized by the EQUIL 2 program and is likely a more accurate measure of stone-forming propensity, although it has not yet gained widespread use.
8. **e. Hyperoxaluria.** Hyperoxaluria has been described in both stone-forming and non-stone-forming patients who have undergone Roux-en-Y gastric bypass surgery, with urinary oxalate levels in some patients exceeding 100 mg/day. A mild decrease in urinary calcium compared with stone-forming control subjects has been described by some investigators but is a less consistent and severe finding.
9. **d. 1,25-dihydroxyvitamin D_3.** It is generally accepted that 1,25-dihydroxyvitamin D_3 is the vitamin D metabolite that is the most potent stimulator of intestinal calcium absorption. The other metabolites, except for calcitonin, are precursors of 1,25-dihydroxyvitamin D_3.
10. **b. Low dietary calcium intake.** Intestinal oxalate absorption is modulated by dietary oxalate and calcium intake and by the presence or absence of *O. formigenes.* In the setting of a high calcium intake, oxalate absorption decreases, and during calcium restriction, oxalate absorption increases because of reduced formation of a soluble calcium oxalate complex and increased availability of oxalate for absorption. *H. pylori,* which can colonize the stomach, has no effect on intestinal oxalate absorption. *O. formigenes*, an oxalate-degrading bacterium, uses oxalate as a substrate in the intestinal lumen, thereby reducing oxalate

absorption. Irritable bowel syndrome, unless it is associated with chronic diarrhea, does not affect intestinal oxalate absorption.

11. **a. Acid–base status**. Citric acid intake has a limited effect on urinary citrate excretion because only a small portion of dietary citrate is excreted into the urine unmetabolized. The majority of absorbed citrate is metabolized to bicarbonate, which is neutralized by the free proton from citric acid, thereby providing no net alkali load that would increase urinary citrate excretion.
12. **d. Primary renal wasting of calcium.** In this condition, the underlying abnormality is a primary renal leak of calcium due to impaired renal tubular calcium reabsorption.
13. **e. Fat malabsorption.** Malabsorption from any cause, including small bowel resection, intrinsic disease, or jejunoileal bypass, increases luminal fatty acids and bile salts. In addition, poorly absorbed bile salts increase colonic permeability to oxalate, further increasing oxalate absorption.
14. **a. Defective ammoniagenesis.** Patients with type 2 diabetes mellitus typically exhibit characteristics of the metabolic syndrome, including insulin resistance. Although peripherally, insulin resistance leads to typical symptoms of diabetes, insulin resistance at the level of the kidney leads to impaired ammoniagenesis, by way of reduced production of ammonia from glutamine and reduced activity of the Na^+/H^+ exchanger in the proximal tubule that is responsible for either the direct transport or trapping of ammonium in the urine. The result is reduced urinary ammonium and low urine pH.
15. **a. Basement membrane of the thin loops of Henle.** In idiopathic calcium oxalate stone formers, Randall plaques have been found to originate in the basement membrane of the thin loops of Henle. From there, they extend through the medullary interstitium to a subepithelial location, where they serve as an anchoring site for calcium oxalate stone formation.
16. **d. Calcium apatite.** Randall plaques are invariably composed of calcium apatite, which serve as an anchoring site onto which calcium oxalate crystals can adhere and grow.

17. **c. Both urinary calcium and oxalate concentrations.** Urinary saturation of calcium oxalate is strongly, positively correlated with urinary calcium and oxalate concentrations. Both contribute equally to urinary saturation of calcium oxalate.
18. **e. Using oxalate as a substrate in the intestine, thereby reducing intestinal oxalate absorption.** *O. formigenes* is an oxalate-degrading bacterium found in the intestinal lumen that uses oxalate as an energy source, thereby reducing luminal oxalate and intestinal oxalate absorption. *Oxalobacter* is not found in urine.
19. **a. *Staphylococcus aureus.*** Although *Proteus* species are most commonly associated with struvite stones, more than 90% of *S. aureus* organisms produce urease and are therefore associated with struvite stone formation.
20. **b. Defective H^+-ATPase in the distal tubule that is unable to excrete excess acid.** A defective H^+-ATPase in the distal tubule has been implicated in the inability to excrete excess acid in the presence of an oral acid load among patients with distal RTA.
21. **c. Xanthine.** Patients with Lesch-Nyhan syndrome suffer from an inherited deficiency of the purine salvage enzyme hypoxanthine-guanine phosphoribosyltransferase, which leads to the accumulation of hypoxanthine, which is ultimately converted to uric acid. Allopurinol inhibits xanthine oxidase, which is responsible for converting hypoxanthine to xanthine and xanthine to uric acid. High doses of allopurinol in these patients lead to the accumulation of hypoxanthine and xanthine, but because xanthine is less soluble in urine than is hypoxanthine, xanthine stones form.
22. **e. Chronic dehydration, intracellular acidosis, and low urinary sodium.** Subjects who abuse laxatives are chronically dehydrated, resulting in intracellular acidosis. In addition, urinary sodium is low from sodium loss as a result of the laxatives. In this environment, urate preferentially complexes with the abundant ammonium rather than sodium and produces ammonium acid urate stones.
23. **c. Complexing calcium in urine.** The primary mechanism of action of citrate is as a complexing agent for calcium, thereby

reducing ionic calcium and urinary saturation of calcium oxalate.

24. **e. Hypokalemia.**
25. **b. Proximal tubule.** The primary defect in type 2 or proximal RTA is a failure of bicarbonate reabsorption in the proximal tubule, leading to excessive urinary bicarbonate excretion and metabolic acidosis.
26. **a. Acidic urine.** Patients with uric acid stones often have prolonged periods of acidity in the urine.
27. **b. Calcium phosphate.** Carbonic anhydrase inhibitors such as acetazolamide and topiramate block reabsorption of bicarbonate in the renal proximal and distal tubules, thereby preventing urinary acidification and inducing a metabolic acidosis. Similar to RTA, carbonic anhydrase inhibition results in the formation of calcium phosphate stones because of the high urine pH, hypercalciuria, and hypocitraturia.
28. **c. Increased calcium excretion.** Placental production of 1,25-dihydroxyvitamin D_3 increases intestinal calcium absorption, further increasing urinary calcium.
29. **e. Uric acid.** Uric acid is thought to promote calcium oxalate stone formation by removing urinary inhibitors, by "salting out" calcium oxalate in solution and by heterogeneous nucleation, or epitaxy. The other substances are all natural inhibitors of stone formation.
30. **a. Arginine.** Renal reabsorption of the four dibasic amino acids—cystine, ornithine, lysine, and arginine—is affected by dysfunction of the heteromeric amino acid transporter located in the renal proximal tubule. Because of the poor solubility of cystine in urine, only cystine stones occur in this condition.

Chapter Review

1. Renal calculi are two to three times more common in men than women, and in this country whites have the highest prevalence. They are uncommon before the age of 20 years, and the peak incidence occurs in the fourth to sixth decades of life.

2. The prevalence and incidence of stone disease is directly correlated with body mass index; patients with high body mass index excrete increased levels of oxalate, uric acid, sodium, and phosphorus, and are more likely to have urinary supersaturation for uric acid. The incidence and prevalence of stone disease has been increasing around the world.
3. Concentration product is the product of the concentrations of the chemical components.
4. Solubility product is the concentration at which precipitation of the components occurs.
5. A solution is saturated when the solubility product is exceeded.
6. When the solubility product is exceeded and precipitation does not occur, the solution is said to be metastable. When precipitation occurs, the concentration at that point is called *formation product.*
7. Magnesium and citrate inhibit crystal aggregation (the former complexes with oxalate and the latter with calcium); nephrocalcin inhibits nucleation, growth, and aggregation; Tamm-Horsfall glycoprotein inhibits aggregation; and osteopontin inhibits crystal growth, nucleation, and aggregation of calcium oxalate crystals.
8. Nanobacteria have been implicated in calcifying nanoparticles and serving as a nidus for stone formation.
9. Most stone-forming salts are found in the urine in a supersaturated state. Inhibitors keep them in solution.
10. The noncrystalline component of stones is called *matrix* and generally accounts for about 2.5% of the weight of the stone. It is composed of mucoproteins, carbohydrates, and urinary inhibitors.
11. Parathyroid hormone increases renal calcium reabsorption and enhances phosphate excretion.
12. Patients with small bowel disease or a history of intestinal resection and an intact colon have an increased oxalate absorption.
13. Calcium absorption occurs primarily in the small intestine at a rate that is dependent on calcium intake.
14. Calcium oxalate accounts for 60% of stones; mixed calcium oxalate and hydroxyapatite, 20%; brushite, 2%; uric acid, 10%; struvite, 10%; and cystine, 1%.

15. Hypercalciuria is the most common abnormality identified in calcium stone formation. Hypercalciuria is defined as a urinary excretion greater than 4 mg/kg/day.
16. Absorptive hypercalciuria is defined as an increased urinary calcium excretion after an oral calcium load and is due to increased intestinal absorption of calcium. Alteration of vitamin D receptors and/or sensitivity has been suggested as the etiology. Renal hypercalciuria is due to impaired renal tubular reabsorption of calcium and leads to secondary hyperparathyroidism.
17. Reabsorptive hypercalciuria is due to hyperparathyroidism. The administration of thiazides to patients with primary hyperparathyroidism exacerbates hypercalcemia. Parathormone-like hormone resulting in hypercalcemia is produced by lung, breast, renal, penile, and head and neck tumors; lymphoma; and myeloma.
18. Hyperoxaluria is defined as greater than 40 mg/day excreted in the urine. Foods that are oxalate rich include nuts, chocolate, brewed tea, spinach, broccoli, strawberries, and rhubarb.
19. Hyperuricosuria is defined as urinary uric acid excretion exceeding 600 mg/day.
20. Hyperuricosuria promotes sodium urate formation, which promotes calcium oxalate stone formation through heterogenous nucleation.
21. Citrate inhibits stone formation by complexing with calcium and thereby preventing spontaneous nucleation of calcium oxalate; it inhibits agglomeration and growth of the crystal, and it enhances the inhibitory effect of Tamm-Horsfall glycoprotein. It prevents heterogeneous nucleation of calcium oxalate by monosodium urate.
22. Hypocitraturia is defined as urinary citrate excretion of less than 320 mg/day.
23. Renal tubular acidosis, type 1 (distal tubular RTA) is characterized by calcium phosphate stone formation, hypercalciuria, hypocitraturia, and an increased urinary pH.
24. Low magnesium levels result in reduced inhibitory activity and are often associated with decreased urinary citrate levels.
25. Cystine stones form due to a defect in the transport of four amino acids: cystine, lysine, ornithine, and arginine. It is inherited as an autosomal recessive and accounts for up to 10% of stones in

children. There are two genes involved in the inheritance of the disease. There are three types based on urine excretion amounts: types A, B, and AB.

26. Stones of infection (struvite stones) are composed of magnesium ammonium phosphate and may contain carbonate apatite; they occur in association with urea-splitting bacteria. Urease-producing pathogens include *Proteus, Klebsiella, Pseudomonas,* and *Staphylococcus.*
27. The cause of stones associated with horseshoe kidneys and ureteropelvic junction obstructions is due to both the anatomic abnormality resulting in stasis and an underlying metabolic abnormality.
28. Medullary sponge kidney is characterized by ectasia of the renal collecting ducts and leads to stones through renal acidifying defects, hypercalciuria, and hypocitraturia.
29. Most stones in pregnancy pass spontaneously.
30. The most important determinate of uric acid stone formation is low urinary pH. Low urinary pH in uric acid stone formers is likely due to impaired ammoniagenesis associated with insulin resistance.
31. Medications that may precipitate as stones include triamterene, silica, indinavir, ephedrine, and ciprofloxacin.
32. Heterogeneous nucleation occurs when microscopic impurities or other constituents in the urine promote nucleation by providing a surface on which the crystal components can grow.
33. Intestinal oxalate absorption is modulated by dietary oxalate and calcium intake and by the presence or absence of *O. formigenes.*
34. Acid–base status determines urinary citrate excretion. Metabolic acidosis reduces citrate excretion.
35. The sarcoid granuloma produces 1,25-dihydroxyvitamin D_3, causing increased intestinal calcium absorption, hypercalcemia, and hypercalciuria.
36. Malabsorption from any cause, including small bowel resection, intrinsic disease, or jejunoileal bypass, increases luminal fatty acids and bile salts. Calcium, which normally complexes with oxalate, forming a soluble complex that is lost in the stool, instead binds to

fatty acids, thereby increasing luminal oxalate available for absorption. In addition, poorly absorbed bile salts increase colonic permeability to oxalate, further increasing oxalate absorption.

37. Type 1 or distal tubule RTA is characterized by an impairment in hydrogen ion secretion. Type 2, or proximal RTA, is characterized by impaired bicarbonate reabsorption in the proximal tubule. Type 4 RTA is common in diabetics with chronic renal damage who demonstrate aldosterone resistance.

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Evaluation and Medical Management of Urinary Lithiasis

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Keywords

evaluation; kidney stone; metabolic; nephrolithiasis; urolithiasis

Questions

1. Each of the following could be considered a high-risk feature for stone formation EXCEPT which?
    a. Diabetes
    b. Sarcoidosis
    c. Parkinson disease
    d. Family history of stones
    e. Inflammatory bowel disease
2. Each of the following is a medical condition associated with calcium nephrolithiasis EXCEPT which?
    a. Hyperparathyroidism
    b. Malignancy
    c. Primary hyperoxaluria
    d. Medullary sponge kidney
    e. Lesch-Nyhan syndrome
3. Standard 24-hour urine testing for stone disease should measure the following EXCEPT which?
    a. Sodium

b. Oxalate
c. pH
d. Creatinine
e. Glucose

4. A 45-year-old male with a history of recurrent calcium oxalate stones returns for counseling regarding a recent 24-hour urine test. A single 24-hour urine collection reveals abnormal elevations in calcium, oxalate, uric acid, sodium, and potassium. Urinary volume is 4.5 L, pH is 5.8, and 24-hour creatinine per kilogram is 47. These findings are most likely explained by:
    a. dietary indiscretion
    b. steroid administration
    c. overcollection
    d. undercollection
    e. contamination
5. Patients with enteric hyperoxaluria are most likely to form stones composed of:
    a. calcium phosphate.
    b. calcium oxalate.
    c. magnesium ammonium phosphate.
    d. uric acid.
    e. cystine.
6. The risk factor most associated with recurrent stone formation in patients with inflammatory bowel disease is:
    a. hyperabsorption of oxalate in the jejunum.
    b. hyperexcretion of calcium from the distal tubule.
    c. diminished citrate absorption in the terminal ileum.
    d. hyperabsorption of calcium in the small bowel.
    e. increased colonic absorption of free oxalate.
7. Hypocitraturia in patients with inflammatory bowel disease or chronic diarrhea syndrome is due to:
    a. persistent bicarbonate losses.
    b. hypokalemia.
    c. metabolic acidosis.
    d. intracellular acidosis.
    e. all of the above.

8. The optimal treatment for patients with enteric hyperoxaluria includes:
    a. calcium supplements, potassium citrate, and increased oral fluid intake.
    b. dietary restriction of oxalate.
    c. thiazides and potassium citrate.
    d. allopurinol.
    e. pyridoxine.
9. The most important factor predisposing patients to gouty diathesis is:
    a. hypercalciuria.
    b. low urinary pH.
    c. hypocitraturia.
    d. low urine volumes.
    e. hyperuricosuria.
10. The most appropriate medical treatment of a patient with uric acid calculi is:
    a. allopurinol.
    b. thiazides.
    c. increased fluids.
    d. dietary calcium restriction.
    e. potassium citrate.
11. A patient with recurrent uric acid calculi is placed on oral medical treatment and returns for follow-up 3 months later. He is noted to have significantly elevated urinary uric acid levels as compared with his first 24-hour urine collection. This finding is due to:
    a. increased production of endogenous uric acid.
    b. failure to avoid high-sodium foods.
    c. increased solubility of uric acid.
    d. increased consumption of red meat.
    e. inhibition of xanthine oxidase.
12. A patient with uric acid calculi is placed on alkali therapy but returns 1 year later having passed two calcium phosphate stones. A repeat 24-hour urine collection demonstrates a urine pH of 7.4, a urinary citrate of 450 mg/day, and a urinary uric acid of 875 mg/day. The most likely cause for recurrent stone formation is:

a. cessation of potassium citrate.
b. increase in oral purine intake.
c. decrease in solubility of uric acid.
d. excess alkalization.
e. increase in saturation of oxalate.

13. A 58-year-old Hispanic woman with a history of recurrent urinary tract infections treated three to four times in the past 18 months is seen by her family physician. At present she is asymptomatic. She has no history of nephrolithiasis. Renal ultrasound demonstrates moderate left hydronephrosis and a large density within the renal pelvis with posterior shadowing. A kidney-ureter-bladder (KUB) view with tomography reveals a poorly opacified dendritic stone in the renal pelvis and lower pole calyces. Prior urine cultures have revealed *Proteus* and *Klebsiella* species. The stone composition of this patient is most likely:
    a. calcium oxalate.
    b. uric acid.
    c. magnesium ammonium phosphate.
    d. cystine.
    e. hydroxyapatite.
14. The most significant factor contributing to stone formation in patients with struvite calculi is:
    a. gouty diathesis.
    b. recurrent urinary tract infections.
    c. family history.
    d. hyperoxaluria.
    e. hypercalciuria.
15. Which of the following bacteria would be the least likely to contribute directly to struvite stone formation?
    a. *Proteus mirabilis*
    b. *Escherichia coli*
    c. *Staphylococcus aureus*
    d. *Klebsiella pneumoniae*
    e. *Pseudomonas aeruginosa*
16. Acetohydroxamic acid (AHA) contributes to reducing infectious stone formation by:

a. reversing associated metabolic defects.
b. preventing recurrent urinary tract infections.
c. alkalization of the urine.
d. irreversibly inhibiting urease.
e. all of the above.

17. A 12-year-old boy is seen for the evaluation of recurrent nephrolithiasis. He has spontaneously passed three stones over the previous 4 years and has recently undergone shock-wave lithotripsy twice without success. He has been treated in the past with an unknown medication, but this was discontinued because the parents believed it to be of no benefit. Urinalysis demonstrates hexagonal crystals. The likely metabolic diagnosis contributing to this patient's recurrent stone formation is:
    a. hypocitraturia.
    b. hyperoxaluria.
    c. low urine volumes.
    d. gouty diathesis.
    e. cystinuria.
18. First-line medical treatment for the prevention of recurrent cystine stones would be aimed at:
    a. urinary acidification.
    b. increasing the solubility of cystine.
    c. decreasing urinary sodium.
    d. decreasing the solubility of cystine.
    e. binding of cystine within the intestines.
19. α-Mercaptopropionylglycine (α-MPG, Thiola Mission Pharmacal Company, San Antonio, TX) may be helpful in the management of cystinuria because it:
    a. acts as a diuretic, further decreasing urinary cystine concentration.
    b. is significantly more effective than D-penicillamine.
    c. can be used as both an oral and intrarenal chemolytic agent.
    d. has equivalent efficacy at increasing solubility with reduced toxicity compared with D-penicillamine.
    e. adequately alkalizes the urine, obviating the need for potassium citrate.

20. Three years after initiating treatment for cystine stones with α-MPG, 800 mg/day, a patient returns with a follow-up 24-hour urine collection demonstrating a significant reduction in cystine excretion from 740 to 250 mg/day. Urine volume is 775 mL/day. He has two additional stones. The reason for recurrent stone formation is:
    a. increased age, thereby exacerbating the disorder.
    b. decreased efficacy of α-MPG.
    c. continued supersaturation of urinary cystine.
    d. continued hypocitraturia.
    e. increased urine acidity.
21. A 19-year-old white woman with a 6-year history of recurrent stone disease is found to have multiple bilateral renal calculi by renal ultrasound during an evaluation for recurrent flank pain. She reports having passed more than 10 stones in the previous 2 years. Review of the renal ultrasound indicates no evidence of hydronephrosis. Her KUB film and tomograms demonstrate five stones on the left and eight stones on the right, all less than 4 mm. She has a strong family history of stones, having three first-degree relatives and two cousins with nephrolithiasis. Urine pH is consistently above 6.8. Stone composition has been mixed calcium phosphate and calcium oxalate. The most definitive test to identify this disorder would demonstrate:
    a. decreased serum parathyroid hormone (PTH) levels.
    b. persistently elevated urine calcium.
    c. inability to reduce the urine pH below 5.3.
    d. normalization of hypercalciuria.
    e. a marked increase in urinary uric acid levels with initiation of treatment.
22. The most appropriate treatment for patients with renal tubular acidosis is:
    a. thiazides.
    b. allopurinol.
    c. sodium alkali.
    d. AHA.
    e. potassium alkali.

23. Renal tubular acidosis may be associated with nephrolithiasis due to:
    a. hypercalciuria and hypocitraturia.
    b. hyperoxaluria and hypercalcemia.
    c. hyperuricosuria.
    d. hypocitraturia with normal urine magnesium.
    e. hypercitraturia and hypercalciuria.
24. Chronic metabolic acidosis may cause:
    a. increased PTH levels.
    b. significantly reduced bone density.
    c. hypercalcemia.
    d. increased intestinal calcium absorption.
    e. all of the above.
25. Which of the following findings would support the diagnosis of renal leak hypercalciuria?
    a. Hypocitraturia
    b. Low urinary pH
    c. Normocalciuria on a calcium-restricted diet
    d. Decreased urinary sodium with thiazide challenge
    e. Low or low-normal radial bone density
26. The primary abnormality in patients with renal leak hypercalciuria is considered to be:
    a. impairment of renal tubular reabsorption of calcium.
    b. excessive mobilization of calcium from bone.
    c. increased 1,25-$(OH)_2$ vitamin D levels.
    d. elevation of serum PTH levels.
    e. hyperabsorption of intestinal calcium.
27. In treating patients with renal leak hypercalciuria, thiazides are effective because they:
    a. bind calcium in the intestinal tract.
    b. cause intracellular volume depletion.
    c. augment calcium reabsorption in the proximal tubule.
    d. directly inhibit calcium absorption.
    e. restore normal serum 1,25-$(OH)_2$ vitamin D levels.

28. Which of the following are potential side effects of treatment with thiazides?
    a. Hypokalemia
    b. Hypocitraturia
    c. Hyperuricemia
    d. Hypomagnesuria
    e. All of the above
29. After 18 months of chlorthalidone treatment, a patient with hypercalciuria is doing well with no further stone formation. However, 8 months later, while still on thiazides, she passed a small stone. The most likely cause of her recurrent stone formation is:
    a. excessive intake of dietary calcium.
    b. inappropriate fluid management.
    c. excessive sodium intake.
    d. thiazide-induced hypocitraturia.
    e. cessation of medications.
30. A patient with hypercalciuria is continued on chlorthalidone and potassium citrate without problems for 18 months and then passes two stones spontaneously. She claims that she is still on her medications. The most likely cause of her continued stone formation is:
    a. excessive calcium intake.
    b. heterogeneous nucleation of calcium oxalate.
    c. high dietary sodium intake.
    d. bone mobilization of calcium.
    e. exacerbation of intestinal calcium absorption.
31. The metabolic abnormality most commonly seen after Roux-en-Y gastric bypass surgery is:
    a. hypercalciuria.
    b. gouty diathesis.
    c. hyperuricosuria.
    d. hyperoxaluria.
    e. hypomagnesemia.

Answers

1. **c. Parkinson disease.** To date, there is no recognized direct association between neurologic disorders and stone formation. The remaining conditions are all potential risk factors for increased likelihood of stone formation and should prompt evaluation with 24-hour urine testing if a stone is diagnosed.
2. **e. Lesch-Nyhan syndrome.** Lesch-Nyhan syndrome is a rare inherited disorder that leads to hyperuricemia and hyperuricosuria with resulting uric acid stone formation. Hyperparathyroidism can lead to calcium stone formation through improper calcium regulation. Malignancy can cause calcium nephrolithiasis through hypercalciuria. Primary hyperoxaluria, though primarily an oxalate disorder, results in calcium oxalate stones most commonly. Medullary sponge kidney may lead to calcium stone formation through urinary stasis and pooling.
3. **e. Glucose.** Based on expert opinion of the guidelines panel, metabolic testing should consist of one or two 24-hour urine collections obtained on a random diet and analyzed at minimum for total volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium and creatinine (Pearle, 2014). Glucose in the urine may be indicative of diabetes, which can be a risk factor for stone formation; however, it is not routinely quantified or measured on 24-hour urine testing for stones.
4. **c. Overcollection.** Normal 24-hour creatinine per kilogram for a male is between 20 and 25. The high value, near 50 in the example, is indicative of excess creatinine in the specimen, most commonly caused by overcollection of urine beyond a 24-hour time period. This is a common occurrence and may happen if a patient does not follow the instructions and provides a 48-hour collection by mistake. The elevated levels of calcium, oxalate, uric acid, sodium, and potassium are thus unreliable, as their values are set based on normal values for a 24-hour time period, not the extended collection time that was likely followed in this study.

5. **b. Calcium oxalate.** Patients with enteric hyperoxaluria are more likely to form calcium oxalate stones owing to increased urinary excretion of oxalate and decreased inhibitory activity from hypocitraturia secondary to chronic metabolic acidosis and hypomagnesuria. In addition, fluid loss from persistent diarrhea from inflammatory bowel disease may cause an extremely concentrated environment that is suitable for stone formation.
6. **e. Increased colonic absorption of free oxalate.** Intestinal hyperabsorption of oxalate in patients with enteric hyperoxaluria is the most significant risk factor leading to recurrent calculus formation. Intestinal transport of oxalate is primarily increased because of the effects of bile salts and fatty acids on the permeability of colonic intestinal mucosa to oxalate. The total amount of oxalate absorbed may also be increased because of an enlarged intraluminal pool of oxalate available for absorption. **Intestinal fat malabsorption characteristic of ileal disease will exaggerate calcium soap formation, limit the amount of "free" calcium to complex to oxalate, and thereby raise the oxalate pool available for absorption.**
7. **e. All of the above. Acid–base status is probably the most important factor in the renal handling of citrate.** Hypokalemia with its induced intracellular acidosis (caused by bicarbonate loss from chronic diarrhea) will reduce urinary citrate by both enhancing renal tubular resorption and reducing the synthesis of citrate. Therefore, in patients with enteric hyperoxaluria in whom bicarbonate loss and hypokalemia both contribute to metabolic acidosis, the hypocitraturia is often profound.
8. **a. Calcium supplements, potassium citrate, and increased oral fluid intake. The initial goals of medical management are to rehydrate and reverse metabolic acidosis.** Hydration is sometimes difficult in some patients because an increase in oral fluids may exacerbate diarrhea. Hydration and potassium citrate will contribute to the reversal of the metabolic acidosis as well as enhance the excretion of citrate to increase its inhibitory effects on stone formation. Calcium supplements will bind excess oxalate within the intestine, thereby reducing intestinal oxalate absorption.

Calcium citrate may offer an ideal calcium supplement in this condition because it should reduce urinary oxalate and increase urinary citrate. Thiazides may worsen metabolic acidosis and hypokalemia through their diuretic effects and renal potassium losses. **Colonic resection may be of benefit in those patients refractory to medical management because the primary site of intestinal absorption of oxalate is the large bowel.**

9. **b. Low urinary pH.** Although low urine volumes and hyperuricosuria contribute to the possibility of uric acid stone formation, the most critical determinant of the crystallization of uric acid remains urinary pH. In addition, uric acid stones may be formed in patients with primary gout with associated severe hyperuricosuria and other secondary causes of purine overproduction, such as myeloproliferative states, glycogen storage disease, and malignancy. Patients with uric acid stones will characteristically have urinary pH lower than the dissociation constant for uric acid (5.5). In fact, many will have a urine pH consistently close to 5. Whereas serum and urine uric acid levels may be elevated in patients with uric acid calculi, the urine pH remains the most cost-effective means of screening for this condition and monitoring therapy.
10. **e. Potassium citrate.** Allopurinol will decrease the production of uric acid by inhibiting xanthine oxidase in the purine metabolic pathway, but it is most effective in patients with extremely elevated levels of uric acid (urinary uric acid >1500 mg/day). In addition, increasing total urine volume will decrease the concentration of uric acid to assist in preventing stone formation. However, raising the urinary pH above the dissociation constant of uric acid is the key to preventing recurrent uric acid stone formation and correcting gouty diathesis. The urine pH should be maintained between 6.0 and 6.5. Thiazides and calcium restriction have limited roles in the medical treatment of patients with uric acid stones.
11. **c. Increased solubility of uric acid.** With adequate alkali therapy, this patient has been able to raise the urine pH above the dissociation constant of uric acid. **The solubility of uric acid is more than 10 times greater at a pH of 7 than at a pH of 5.**

Therefore patients may initially present with low/normal 24-hour urinary uric acid levels because the uric acid will precipitate out of solution in the acid urinary environment. Once the urine has been alkalized, all of the uric acid will come back into solution, causing a significant increase in the measured urinary uric acid.

12. **d. Excess alkalization. Excessive alkalization with urinary pH values above 7.0 may result in calcium phosphate stone formation.** Alkali therapy with potassium citrate should aim to keep the urinary pH between 6.5 and 7.0 when patients with gouty diathesis are being treated.
13. **c. Magnesium ammonium phosphate.** Ascending urinary tract infections with urea-splitting organisms such as *Proteus* species will metabolize urea to ammonia. Ammoniuria—in conjunction with a matrix composed of organic compounds, carbonate apatite, inflammatory cells, and bacteria—results in the rapid formation of an "infectious" calculus, eventually progressing into a mineralized, dense stone. **Bacteria trapped within the stone perpetuate the recurrent urinary tract infections, and further stone formation eventually develops into the classic staghorn calculus.**
14. **b. Recurrent urinary tract infections.** Etiologic factors involved with infection calculi include a history of recurrent urinary tract infections and potential anatomic or physiologic abnormalities. It is important to remember that these patients may also have underlying metabolic disorders such as hypercalciuria, which could contribute to the stone formation. These disorders are most commonly found in patients with mixed stone composition (i.e., struvite and calcium calculi). A comprehensive metabolic evaluation is warranted in these patients.
15. **b. *Escherichia coli.*** Of the listed bacteria, *E. coli* is the only one that does not commonly express urease activity, the enzyme responsible for struvite stone formation in a majority of cases.
16. **d. Irreversibly inhibiting urease.** AHA, a competitive inhibitor of the bacterial enzyme urease, will reduce the urinary saturation of struvite and retard stone formation. When given at a dose of 250 mg orally three times a day, this medication can prevent the development of new stones and inhibit the growth of existing

stones in patients with chronic urea-splitting infections. AHA can also cause dissolution of small stones. However, up to 30% of patients will experience minor side effects including headache, nausea, vomiting, anemia, rash, and alopecia. In addition, 15% of patients have developed deep venous thrombosis while on long-term treatment. Therefore careful monitoring is required when this medication us being used.

17. **e. Cystinuria. Cystinuria is a complex autosomal recessive disorder of amino acid transport involving cystine, ornithine, lysine, and arginine (COLA). Supersaturation of the urine will occur in patients with the homozygous state. Therefore it is unusual to see a family history with cystine stones, and the age at onset is often in the first or second decade.**
18. **b. Increasing the solubility of cystine.** Increasing the solubility of cystine is the mainstay of treating this disorder. Therefore medical therapy is aimed at dissociating cystine into cysteine, which is 200 times more soluble than cystine. Solubility increases dramatically when this disulfide exchange occurs, effectively preventing further stone formation.
19. **d. Has equivalent efficacy at increasing solubility with reduced toxicity compared with d-penicillamine.** D-Penicillamine and α-MPG are equally effective in their ability to decrease urinary cystine levels. However, studies have demonstrated that α-MPG is significantly less toxic than D-penicillamine. Moreover, the side effects that may occur with α-MPG are also less severe. However, if a patient has been doing well on D-penicillamine with no significant complications, there is no need to switch medications.
20. **c. Continued supersaturation of urinary cystine.** The primary goal of medical therapy is to reduce the urinary cystine concentration below the solubility limit of 200 to 250 mg/L of urine. Because many of these patients present at a young age, compliance may be difficult. Even though this patient's cystine excretion has been reduced to 250 mg/day by the α-MPG therapy, his cystine concentration remains greater than 300 mg/L. Therefore a combination of medication along with an increased urine output is essential to reduce the urinary cystine concentration. Long-term

follow-up is necessary to ensure urinary cystine beneath the saturation concentration.

21. **c. Inability to reduce the urine pH below 5.3.** Renal tubular acidosis is a clinical syndrome of chronic metabolic acidosis resulting from renal tubular abnormalities while glomerular filtration is relatively well preserved. Although patients may present with many different symptoms and physical findings, renal stone formation is a well-recognized manifestation of distal renal tubular acidosis (dRTA). Patients with the incomplete form of dRTA are not persistently acidemic despite their inability to lower urinary pH with an acid load. These patients are able to compensate for their acidification defect and remain in acid–base balance by increasing ammonia synthesis and ammonium excretion as a buffering mechanism. The initial identification of incomplete dRTA is often a chance finding. Many of these patients will present with recurrent nephrolithiasis or may be referred for evaluation after the discovery of nephrocalcinosis after routine abdominal radiographs. Most patients will have normal serum electrolytes, yet they will have a high-normal urine pH along with significant hypocitraturia. The diagnosis of incomplete dRTA can be confirmed by inadequate urinary acidification after an ammonium chloride loading test.
22. **e. Potassium alkali.** In the past, sodium alkali has been the treatment of choice for chronic therapy in patients with dRTA. It was given in the form of either sodium bicarbonate or Shohl solution (a combination of sodium citrate and citric acid). Although sodium alkali is beneficial in correcting the acidosis, excess sodium may be detrimental to calcium metabolism, especially with respect to nephrolithiasis. **Sodium alkali therapy has been complicated by the development of calcium stones (calcium phosphate or calcium oxalate), especially when the urinary pH is above 7.** Potassium citrate has been shown to reduce the excretion of urinary calcium, whereas sodium alkali has no effect on urinary calcium. Therefore potassium alkali, usually in the form of potassium citrate (POLYCITRA-K, ALVA Pharmaceuticals, Mountain View, CA; Urocit-K, Mission Pharmacal Company, Mountain View, CA), is the recommended first-line therapy.

23. **a. Hypercalciuria and hypocitraturia.** Hypocitraturia, commonly seen in patients with dRTA, promotes the formation of nephrolithiasis due to reduced inhibitory action of urinary citrate. In addition, hypercalciuria will occur due to mobilization of calcium from bone and impaired renal tubular absorption of calcium, both as a result of chronic acidosis.
24. **b. Significantly reduced bone density.** It is well established that metabolic acidosis may cause a negative calcium balance as a result of impaired renal tubular reabsorption of calcium in the proximal tubule, leading to excessive renal loss of calcium. In addition, intestinal calcium absorption is diminished in patients with persistent acidosis. Slow dissolution of bone mineral can also be identified as calcium and phosphate act as buffering mechanisms to correct the acidosis. Chronic acidosis has been cited as a major factor in the genesis of bone disease.
25. **e. Low or low-normal radial bone density.** Patients with renal hypercalciuria may display a low or low-normal radial bone density. The diminished bone density is a result of the secondary hyperparathyroidism, which causes stimulation of PTH and subsequent production of 1,25-$(OH)_2$ vitamin D. Both PTH and vitamin D will act on bone to mobilize calcium and cause a loss in bone density. Calcium restriction has no effect in managing renal hypercalciuria.
26. **a. Impairment of renal tubular reabsorption of calcium. The primary abnormality in renal hypercalciuria is an impairment in proximal renal tubular calcium reabsorption.** This urinary calcium wasting and subsequent reduction in serum calcium concentration stimulates the production of PTH. As a result, vitamin D synthesis in the kidney is stimulated. Both PTH and vitamin D will increase bone resorption and absorption of intestinal calcium, increasing the circulating concentration and filtered load of calcium. This often causes significant hypercalciuria. Unlike primary hyperparathyroidism, serum calcium is normal and the state of hyperparathyroidism is secondary.

27. **c. Augment calcium reabsorption in the proximal tubule.** Thiazide is the primary medical treatment of renal hypercalciuria and has been shown to correct the renal leak of calcium by augmenting calcium reabsorption in the distal tubule. In addition, thiazides cause extracellular volume depletion, thereby stimulating proximal tubular reabsorption of calcium. A positive calcium balance ensues, with correction of the secondary hyperparathyroidism.
28. **e. All of the above. Thiazide diuretics can cause hypokalemia. Symptoms of hypokalemia include muscle cramps and weakness. Consideration should be given to starting patients concurrently on potassium supplementation with potassium citrate, as patients can also become hypocitraturic.** It is reasonable to check a basic metabolic panel 1 to 2 weeks after initiating a thiazide to monitor potassium levels. Thiazides can also cause low urinary citrate and magnesium. In addition, they can lead to impaired carbohydrate metabolism and hyperuricemia. A small percentage of patients may also have sexual side effects including decreased libido or erectile dysfunction.
29. **d. Thiazide-induced hypocitraturia.** Intracellular acidosis resulting from thiazide-induced hypokalemia will augment renal tubular reabsorption of citrate with resultant hypocitraturia. The reduction in the inhibitory effects of hypocitraturia may promote further stone formation. Therefore potassium repletion is necessary if long-term thiazide treatment is anticipated. Our potassium supplement of choice is potassium citrate, in either pill or liquid preparations.
30. **c. High dietary sodium intake.** A high dietary sodium intake has two deleterious effects in this case. An excess sodium load will inhibit reabsorption of calcium in the proximal tubule, thereby causing hypercalciuria. Moreover, sodium will block the hypocalciuric action of thiazides. Therefore patients placed on thiazide diuretics for management of hypercalciuria should also be placed on a dietary sodium restriction.
31. **d. Hyperoxaluria.** Roux-en-Y gastric bypass has been shown to lead to significantly increased urinary oxalate. Other metabolic

abnormalities seen in patients with Roux-en-Y gastric bypass include low urine volumes and hypocitraturia. In contrast, restrictive bariatric surgery—such as gastric band or sleeve—does not lead to hyperoxaluria. These patients have low urine volumes as their primary metabolic abnormality.

Chapter Review

1. First-time stone formers are at a 50% risk for recurrence. Males have both a higher incidence of calculi and a higher recurrence rate.
2. Infectious calculi may contain large quantities of endotoxin.
3. A complete urine collection is confirmed by the 24-hour excretion of creatinine. On average 1 mg/kg per hour is excreted.
4. As the phosphate content of the stone increases from calcium oxalate to calcium oxalate-calcium apatite to calcium apatite, the incidence of renal tubular acidosis increases from 5% to 39% and the incidence of primary hyperthyroidism increases from 2% to 10%. Thus higher phosphate content in stones correlates with an increased incidence of renal tubular acidosis and primary hyperparathyroidism.
5. Obesity increases the risk of nephrolithiasis.
6. Hypercalciuria not associated with hypercalcemia may be subdivided into (1) excess gastrointestinal absorption, (2) renal tubular leak, or (3) normocalcemic hyperparathyroidism.
7. Patients with hyperuricosuria have increased calcium oxalate urolithiasis due to heterogeneous nucleation.
8. Increased protein intake increases the likelihood of renal stones due to increased urinary calcium, oxalate, and uric acid excretion. Moderate calcium ingestion and a reduced sodium diet, when combined with animal protein restriction, reduce calcium stone episodes by approximately 50%.
9. Thiazides may unmask primary hyperparathyroidism by causing a marked rise in serum calcium. They also cause hypocitraturia.
10. Indinavir stones may not be visible on CT.

11. Furosemide-induced nephrolithiasis should always be considered in neonates who develop nephrolithiasis.
12. Children with stones should always be worked up because inborn errors of metabolism may be responsible for the stones in a significant number of patients. The inborn errors of metabolism most commonly found in this circumstance include cystinuria, renal tubular acidosis, and primary hyperoxaluria.
13. Patients with diabetes mellitus may have altered ammonium metabolism resulting in acidic urine that predisposes them to calcium oxalate and/or uric acid calculi.
14. Sulfate content in a 24-hour urine collection is an indication of the amount of protein intake. Protein intake increases calcium, oxalate, and uric acid excretion.
15. Intestinal fat malabsorption characteristic of ileal disease will exaggerate calcium soap formation, limit the amount of "free" calcium to complex to oxalate, and thereby raise the oxalate pool available for absorption.
16. Acid–base status is probably the most important factor in the renal handling of citrate.
17. The primary site of intestinal absorption of oxalate is the large bowel.
18. The solubility of uric acid is more than 10 times greater at a pH of 7 than at a pH of 5.
19. Sodium alkali therapy has been complicated by the development of calcium stones (calcium phosphate or calcium oxalate), especially when the urinary pH is above 7.

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Strategies for Nonmedical Management of Upper Urinary Tract Calculi

David A. Leavitt, Jean J.M.C.H. de la Rosette, and David M. Hoenig

Keywords

calculi; kidney stones; nephrolithiasis; surgery; surgical management; surgical treatment; ureteral stones

Questions

1. The best predictor of post–percutaneous nephrolithotomy (PNL) urosepsis is:
    a. preoperative bladder urine culture.
    b. intraoperative bladder urine culture.
    c. stone culture.
    d. preoperative blood culture.
    e. intraoperative blood culture.
2. What is the risk of mortality from an untreated struvite staghorn stone?
    a. Less than 10%
    b. 10% to 30%
    c. 30% to 50%
    d. 50% to 70%
    e. Greater than 70%

3. The increased risk of residual fragments after extracorporeal shockwave lithotripsy (SWL) of large-volume calculi is of particular importance for patients with stones composed of:
    a. brushite.
    b. uric acid.
    c. struvite.
    d. calcium oxalate monohydrate.
    e. calcium oxalate dihydrate.
4. What is the single most important factor when choosing among SWL, ureteroscopic stone removal, and PNL for renal calculi?
    a. Stone composition
    b. Stone location
    c. Anatomic abnormalities
    d. Stone burden
    e. Body habitus
5. What is the preferred treatment for a known brushite stone former harboring a lower pole renal calculus 25 mm in diameter?
    a. SWL
    b. SWL with ureteral stenting
    c. Flexible ureteroscopy with holmium laser lithotripsy
    d. PNL
    e. Laparoscopic pyelolithotomy
6. What is the preferred initial treatment for staghorn calculi?
    a. SWL with ureteral stenting
    b. Flexible ureteroscopy with holmium laser lithotripsy
    c. PNL
    d. Extended pyelolithotomy
    e. Anatrophic nephrolithotomy
7. Which of the following is the most difficult stone composition to fragment with SWL?
    a. Calcium oxalate dihydrate
    b. Calcium oxalate monohydrate
    c. Struvite
    d. Hydroxyapatite
    e. Uric acid

8. What is the preferred treatment approach for a symptomatic 1.5-cm stone in a lower pole calyceal diverticulum?
    a. SWL
    b. Flexible ureteroscopy
    c. PNL
    d. PNL with fulguration of the diverticulum
    e. Laparoscopic diverticulectomy
9. What is the preferred initial treatment for a 10-mm stone in the renal pelvis of a horseshoe kidney with minimal hydronephrosis?
    a. SWL
    b. Flexible ureteroscopy
    c. PNL
    d. Laparoscopic pyelolithotomy
    e. Symphysiotomy with pyelolithotomy
10. What is the preferred treatment approach for a 10-mm renal calculus in a patient who weighs 375 lb?
    a. SWL
    b. Flexible ureteroscopy
    c. PNL
    d. SWL using the “blast path” technique
    e. Open surgery
11. What is the preferred treatment option for a patient with a symptomatic 1.5-cm renal calculus and a coagulopathy?
    a. SWL
    b. SWL after administration of fresh-frozen plasma
    c. Indwelling ureteral stent
    d. Flexible ureteroscopy
    e. PNL
12. Residual fragments after SWL have been associated with which of the following?
    a. Hypertension
    b. An increased rate of recurrent stones
    c. A decreased rate of recurrent stones
    d. Perinephric hematomas
    e. Hematuria

13. What is the most sensitive test for identifying residual fragments after PNL?
    a. Nephrotomography
    b. Magnetic resonance imaging (MRI)
    c. Ultrasonography
    d. Noncontrast computed tomography (CT)
    e. Contrast-enhanced CT
14. Factors affecting the probability of spontaneous passage of ureteral calculi include all of the following EXCEPT:
    a. stone size.
    b. stone location at presentation.
    c. stone composition.
    d. degree of hydronephrosis.
    e. duration of symptoms.
15. Irreversible loss of renal function can occur within what time period when a completely obstructing ureteral stone is present?
    a. 1 week
    b. 2 to 4 weeks
    c. 4 to 6 weeks
    d. More than 6 weeks
    e. 3 months
16. A first-time stone former is diagnosed with a 4-mm proximal ureteral calculus. The best initial management is:
    a. ureteroscopic laser lithotripsy.
    b. ureteral stent placement.
    c. SWL.
    d. expectant management.
    e. SWL with ureteral stent placement.
17. Large-volume matrix calculi, which form as a consequence of urinary tract infection, are:
    a. effectively fragmented with SWL.
    b. best approached in a ureteroscopic fashion.
    c. generally sterile.
    d. radiopaque and well visualized on plain radiographic studies.
    e. most efficiently treated with PNL.

18. Ureteral stent placement when SWL is performed for ureteral stones is appropriate for all of the following reasons EXCEPT:
    a. solitary kidney.
    b. relief of severe symptoms.
    c. enhancement of stone fragmentation.
    d. relief of obstruction.
    e. aid in localization of difficult-to-visualize stones.
19. The preferred single agent for medical expulsive therapy for distal ureteral calculi is:
    a. nifedipine.
    b. tamsulosin.
    c. Solu-Medrol.
    d. ibuprofen.
    e. terazosin.
20. The treatment modality associated with the greatest stone-free rates and the least morbidity for patients with distal ureteral stones of any size is:
    a. PNL.
    b. SWL.
    c. ureteroscopy.
    d. open ureterolithotomy.
    e. laparoscopic ureterolithotomy.

Answers

1. **c. Stone culture. The best predictor of post-PNL urosepsis is stone culture or renal pelvic urine culture results.**
2. **b. 10% to 30%. The 10-year mortality rate of untreated staghorn stones was 28%, versus 7.2% in patients treated with surgery**.
3. **c. Struvite.** Struvite stones must be removed completely to minimize the risk of continued urea-splitting bacteriuria.
4. **d. Stone burden**. **Stone burden (size and number) is perhaps the single most important factor in deciding the appropriate treatment modality for a patient with kidney calculi.**

5. **d. PNL.** The Lower Pole Stone Study Group compared ureteroscopy and PNL for patients with 10- to 25-mm lower pole stones and found a significant difference in stone clearance, with only 40% of the ureteroscopic cohort stone free at 3 months versus 76% of the PNL cohort.
6. **c. PNL**. **The management of staghorn stones with a combined approach must be viewed as primarily percutaneous, with SWL being used only as adjunct to minimize the number of accesses required.**
7. **b. Calcium oxalate monohydrate**. **Cystine and brushite are the stones most resistant to SWL, followed by calcium oxalate monohydrate. Next, in descending order, are hydroxyapatite, struvite, calcium oxalate dihydrate, and uric acid stones.**
8. **d. PNL with fulguration of the diverticulum. The percutaneous approach for the management of patients with calyceal diverticular stones provides the patient with the best chance of becoming stone and symptom free.** Fulguration of the diverticulum will reduce the risk of recurrence of the diverticulum.
9. **a. SWL.** SWL can achieve satisfactory results in properly selected patients, such as those with small stones (<1.5 cm) in the presence of normal urinary drainage. For larger stones, or when there is evidence of poor urinary drainage, PNL should be used as the primary approach.
10. **b. Flexible ureteroscopy.** Retrograde ureteroscopic intrarenal surgery may be the preferred modality of treatment for morbidly obese patients when the stone burden is not excessively large.
11. **d. Flexible ureteroscopy. When anticoagulation cannot be temporarily discontinued, the use of ureteroscopy in combination with holmium laser lithotripsy is preferred.** One study reported that even when patients' coagulopathies were not fully corrected, the stones could be successfully treated with no increase in hemorrhagic complications.
12. **b. An increased rate of recurrent stones.** At follow-up (1.6 to 85.4 months), 43% of the patients with residual fragments had a significant symptomatic episode or required intervention.

13. **d. Noncontrast computed tomography (CT).** Although flexible nephroscopy is often considered the gold standard for assessing residual stones after PNL, the routine use of flexible nephroscopy has been challenged by studies showing the high sensitivity of noncontrast CT in detecting residual stones. Noncontrast CT had 100% sensitivity for detecting residual stones after PNL in 36 patients evaluated by both CT and flexible nephroscopy.
14. **c. Stone composition.** One study analyzed 75 patients with ureteral calculi and found that the interval to stone passage was highly variable and dependent on stone size, location, and side. Stones that were smaller, more distal, and on the right side were more likely to pass spontaneously. In another study, duration of symptoms before presentation was the most influential factor, followed by the degree of hydronephrosis.
15. **b. 2 to 4 weeks.** Even with complete ureteral obstruction, irreversible loss of renal function does not occur for more than 2 weeks but can progress to total renal unit loss at up to 6 weeks.
16. **d. Expectant management**. **The majority of ureteral stones less than 5 mm will pass spontaneously and therefore can be treated with expectant management**.
17. **e. Most efficiently treated with PNL. Matrix stones are most effectively treated with PNL.** SWL is usually ineffective because of the stone's gelatinous nature, and ureteroscopy may be compromised by the large volume of stone material present.
18. **c. Enhancement of stone fragmentation.** Although early reports supported the routine use of a ureteral stent to bypass ureteral stones before SWL, data analyzed by the American Urological Association Ureteral Calculi Guidelines Panel showed no improvement in fragmentation with stenting, and therefore routine stent placement before SWL was discouraged. However, ureteral stent placement is appropriate for other indications, such as management of pain, relief of obstruction, and stones that are difficult to visualize, and is mandatory in a patient who has a solitary obstructed kidney.
19. **b. Tamsulosin. Tamsulosin, a selective α-adrenergic blocker, is the preferred agent for medical expulsive therapy, owing to its**

reported efficacy and superior side effect profile.

20. **c. Ureteroscopy.** The stone-free rate for distal ureteral stones approached with ureteroscopy was 91% in the American Urological Association/European Association of Urology Ureteral Stones Guidelines document, an outcome superior to SWL.

Chapter Review

1. Determinants of poor stone clearance rates after SWL include large renal calculi, lower pole or obstructed portions of the collecting system, very hard stones, and obesity.
2. Most calyceal stones in the absence of intervention are likely to increase in size and cause pain or infection.
3. If left untreated, staghorn calculi are likely to be associated with a progressive decrease in renal function.
4. Patients with cystinuria are more likely to have decreased renal function than other stone formers.
5. There is a linear correlation between Hounsfield units (density) and success of SWL.
6. Complete stone clearance from lower pole calyces is less likely than from other calyces for reasons that are not totally clear.
7. Spontaneous passage of a distal ureteral stone is more likely than that of a proximal ureteral stone.
8. For patients with proximal ureteral stones, there is no difference between treating the stone in situ or pushing it back into the renal pelvis when utilizing SWL.
9. In humans, there is no clear time threshold for irreversible damage in complete ureteral obstruction. It is clear, however, that patients with compromised vasculature, decreased renal reserve, poor nutrition, and other comorbid diseases such as diabetes tolerate obstruction less well than do patients with normal kidneys. Most agree that a significantly obstructing stone should not be allowed to persist beyond 2 to 4 weeks.
10. Untreated staghorn calculi are associated with recurrent urinary tract infections, urosepsis, renal function deterioration, and increased

mortality.

11. Renal stones less than 1 cm are best treated with SWL or ureterorenoscopic extraction; stones between 1 and 2 cm that are not in the lower pole are best treated with SWL or PNL; stones greater than 2 cm are best treated with PNL.
12. Matrix renal stones are best removed with PNL.
13. Alpha blockers increase the rate of spontaneous passage of ureteral stones. Tamsulosin, a selective α-adrenergic blocker, is the preferred agent for medical expulsive therapy.
14. Bacteria may reside deep in the stone and may be impossible to eradicate without complete stone removal.
15. Medications responsible for producing stones include ephedrine, indinavir, triamterene, magnesium trisilicate, ciprofloxacin, and sulfa drugs.
16. The best predictor of post-PNL urosepsis is stone culture or renal pelvic urine culture results.
17. Stone burden (size and number) is perhaps the single most important factor in deciding the appropriate treatment modality for a patient with kidney calculi.
18. Cystine and brushite are the stones most resistant to SWL, followed by calcium oxalate monohydrate.
19. The percutaneous approach for the management of patients with calyceal diverticular stones provides the patient with the best chance of becoming stone and symptom free.
20. When anticoagulation cannot be temporarily discontinued, the use of ureteroscopy in combination with holmium laser lithotripsy is preferred.
21. The majority of ureteral stones less than 5 mm will pass spontaneously.
22. Pregnant women who require admission and require treatment for renal colic have a greater risk of preterm delivery.
23. Pregnancy induces a state of absorptive hypercalciuria and mild hyperuricosuria that is offset by increased excretion of urinary inhibitors such as citrate and magnesium, as well as increased urinary output.

24. SWL is now known to induce acute structural changes in the treated kidney in most, if not all, patients. The insult is primarily a vascular injury.
25. Uncorrected coagulopathy and an active, untreated urinary tract infection are two absolute contraindications to PNL.
26. Physiologic solutions should be used for irrigation during PNL to minimize the risk of dilutional hyponatremia in the event of large-volume extravasation.